



# GOSFORD HIGH SCHOOL

2011  
HIGHER SCHOOL CERTIFICATE

## EXTENSION 2 MATHEMATICS

### ASSESSMENT TASK 2

#### General Instructions:

- Reading time: 5 minutes.
- Working time: 90 minutes
- Write using black or blue pen.
- Board-approved calculators may be used.
- Each question should be started on a new page.
- All necessary working should be shown in every question.

Total marks: - 63

Attempt all Questions 1- 4.

#### Question 1:

(a) If  $z = 2 - 3i$  and  $\omega = 1 - i$  find in the form  $a + ib$

(i)  $z + \omega$  (1)

(ii)  $z\omega$  (1)

(iii)  $\frac{z}{\omega}$  (2)

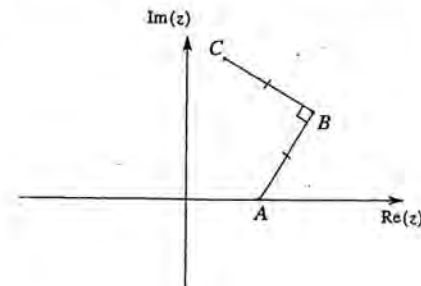
(b) (i) Express  $1 - \sqrt{3}i$  in modulus - argument form. (1)

(ii) Hence evaluate  $(1 - \sqrt{3}i)^4$  in the form  $a + ib$ . (2)

(c) Sketch the region in the complex plane where:

$|z + 2 - i| \leq 3$  and  $-\frac{\pi}{3} \leq \arg z \leq \frac{\pi}{4}$  (2)

(d) The diagram below shows the fixed points A, B & C in the Argand plane, where  $AB = BC$  and  $\angle ABC = \frac{\pi}{2}$ . The point A represents the complex number  $z_1 = 2$  and the point B represents the complex number  $z_2 = 3 + \sqrt{5}i$ .



(i) Find the complex number  $z_3$  represented by the point C (2)

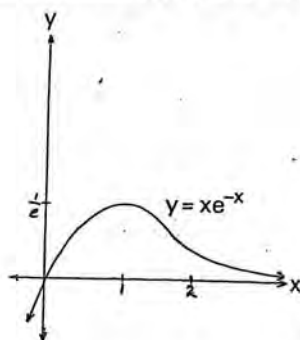
(ii) D is the point in the Argand plane such that ABCD is a square. Find the complex number  $z_4$  represented by the point D. (1)

(e) Find, in Cartesian form, the equation of the locus of the point z if

$\operatorname{Re}\left(\frac{z-2}{z}\right) = 0$ . (4)

**Question 2:**

- (a) The function defined by  $f(x) = xe^{-x}$  is drawn below.



Draw separate one third page sketches showing the important features of each of the following.

- (i)  $y = |f(x)|$  (2)
  - (ii)  $y = f(|x|)$  (2)
  - (iii)  $y = \frac{1}{f(x)}$  (2)
  - (iv)  $y = \sqrt{f(x)}$  (2)
  - (v)  $|y| = f(x)$  (2)
- (b)
- (i) If  $y^2 = x^3(x-2)$  use implicit differentiation to show that for  $y > 0$ ,  $\frac{dy}{dx} = (2x-3)\sqrt{\frac{x}{x-2}}$ . (2)
  - (ii) Draw a neat sketch of  $y = x^3(x-2)$  clearly showing the coordinates of the stationary point between  $x = 0$  &  $x = 2$ . (3)
  - (iii) On the same number plane sketch the graph of  $y^2 = x^3(x-2)$  showing all important features. (2)

**Question 3:**

- (a)
  - (i) Prove that if a polynomial  $P(x)$  has a root  $\alpha$  of multiplicity  $r$ , then  $P'(x)$  has a root  $\alpha$  of multiplicity  $r-1$ . (2)
  - (ii) If  $P(x) = x^4 - 5x^3 + 6x^2 + 4x - 8$  has a root of multiplicity 3, find all the roots. (3)
- (b) It is given that  $1+i$  is a root of  $P(z) = 2z^3 - 3z^2 + rz + s$  where  $r$  &  $s$  are real numbers.
  - (i) Explain why  $1-i$  is also a root of  $P(z)$ . (1)
  - (ii) Factorise  $P(z)$  over the Real field  $\mathbb{R}$ . (3)
- (c) Factorise  $x^3 - 3x^2 + 4x - 2$  over the complex field  $\mathbb{C}$ . (3)
- (d) Two of the roots of  $3x^4 - 10x^3 + px^2 + qx - 12$  are reciprocals while the other two roots are opposites of each other. Find the roots and evaluate  $p$  &  $q$ . (4)
- (e) If  $z = \cos\theta + i\sin\theta$ , show that  $z^n + z^{-n} = 2\cos n\theta$ . Hence show that  $32\cos^5\theta = 2\cos 5\theta + 10\cos 3\theta + 20\cos\theta$ . (4)

**Question 4:**

- (a) Find  $\int \frac{e^x}{\sqrt{1-e^{2x}}} dx$ . (2)
- (b) Use the substitution  $u = \sqrt{2x-1}$  to evaluate  $\int_1^2 \frac{dx}{x\sqrt{2x-1}}$ . (4)
- (c)
  - (i) Find real numbers  $a$  and  $b$  such that:
 
$$\frac{7}{2x^2+5x-3} = \frac{a}{x+3} + \frac{b}{2x-1}$$
 (3)
  - (ii) Hence find  $\int \frac{7}{2x^2+5x-3} dx$  (1)

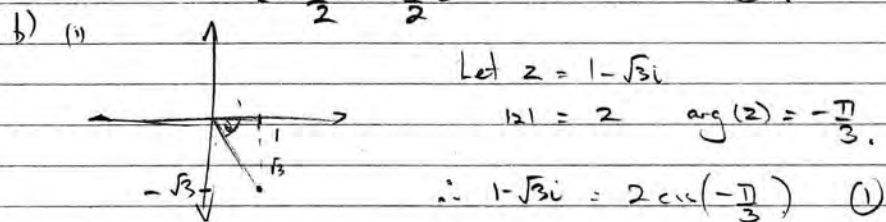
# SOLUTIONS

QUESTION 1.

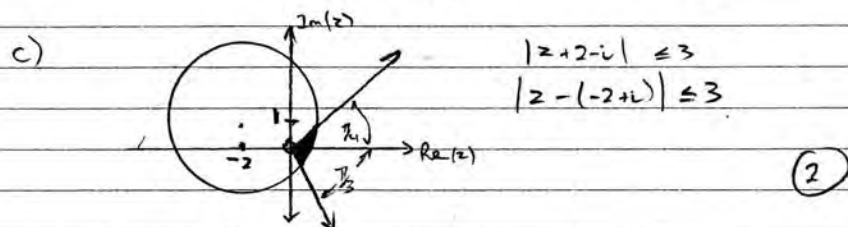
a) (i)  $z + w = 2 - 3i + 1 - i$   
 $= 3 - 4i$  (1)

(ii)  $zw = (2 - 3i)(1 - i)$   
 $= 2 - 2i - 3i + 3i^2$   
 $= -1 - 5i$  (1)

(iii)  $\frac{z}{w} = \frac{(2 - 3i) \times (1 + i)}{(1 - i)(1 + i)}$   
 $= \frac{2 + 2i - 3i - 3i^2}{1 - i^2}$   
 $= \frac{5 - i}{2}$  (2)



(ii)  $(1 - \sqrt{3}i)^4 = \left[2 \operatorname{cis}\left(-\frac{\pi}{3}\right)\right]^4$   
 $= 16 \operatorname{cis}\left(-\frac{4\pi}{3}\right)$   
 $= 16 \operatorname{cis} \frac{2\pi}{3}$   
 $= 16 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$   
 $= 16 \left(-\frac{1}{2} + i \cdot \frac{\sqrt{3}}{2}\right)$   
 $= -8 + 8\sqrt{3}i$  (2)



d) (i)  $\vec{OC} = \vec{OA} + \vec{AB} + \vec{BC}$   
 $= 2 + 1 + \sqrt{5}i + i(1 + \sqrt{5}i)$   
 $= 2 + 1 + \sqrt{5}i + i - \sqrt{5}$   
 $= (3 - \sqrt{5}) + (1 + \sqrt{5})i$  (2)

$\therefore z_3 = (3 - \sqrt{5}) + (1 + \sqrt{5})i$   
 (ii)  $\vec{OD} = \vec{OA} + \vec{BC}$   
 $= 2 + i - \sqrt{5}$   
 $= (2 - \sqrt{5}) + i$   
 $\therefore z_4 = (2 - \sqrt{5}) + i$  (1)

e) Let  $z = x + iy$ .

$\frac{z-2}{z} = \frac{z+iy-2}{x+iy} \times \frac{x-iy}{x-iy}$   
 $= \frac{x^2 - iyx + ixy - i^2y^2 - 2x + 2iy}{x^2 + y^2}$   
 $= \frac{x^2 - 2x + y^2}{x^2 + y^2} + \frac{i \cdot 2y}{x^2 + y^2}$

If  $\operatorname{Re}\left(\frac{z-2}{z}\right) = 0$ ,

$\frac{x^2 - 2x + y^2}{x^2 + y^2} = 0$

$x^2 - 2x + y^2 = 0, \quad x \neq 0, y \neq 0$  (4)

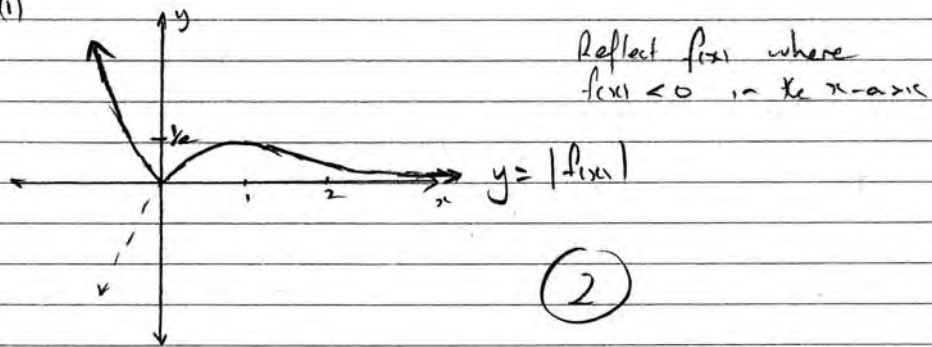
$x^2 - 2x + 1 + y^2 = 1$   
 $(x-1)^2 + y^2 = 1$

This is a circle centre (1,0) & radius 1 unit excluding the origin.

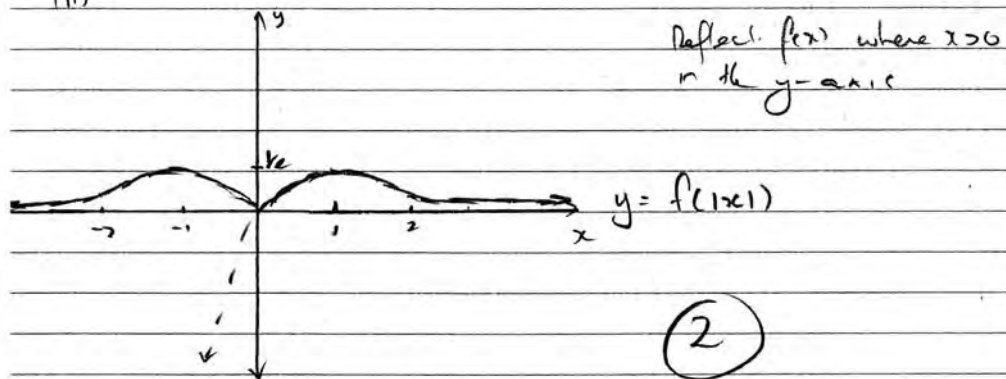
## QUESTION 2.

a)

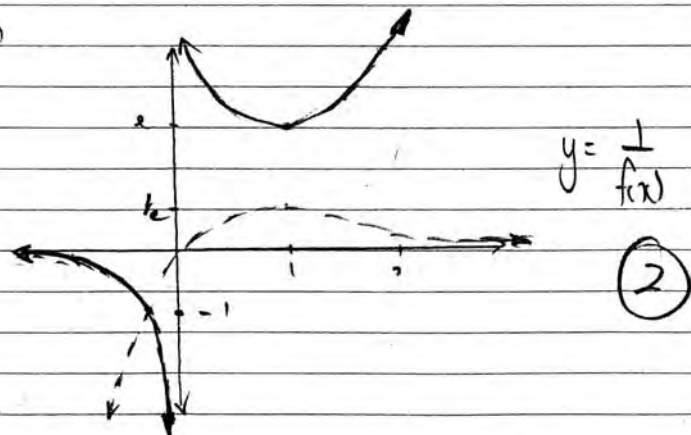
(i)



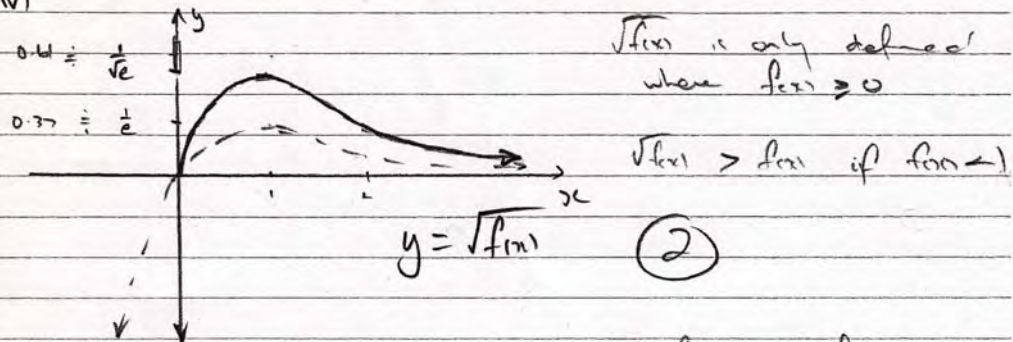
(ii)



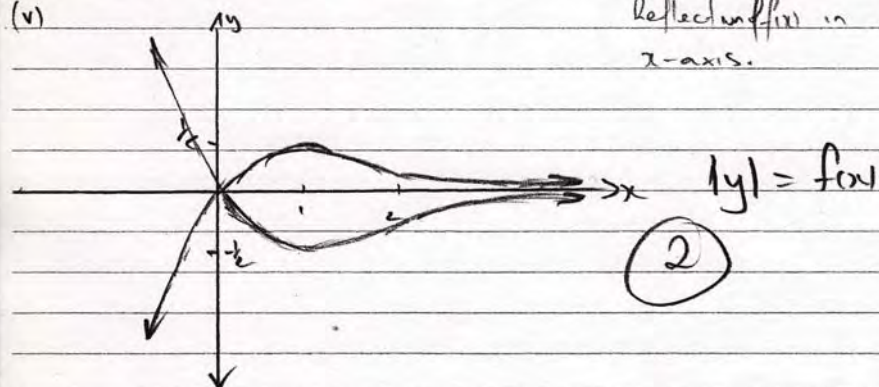
(iii)



(iv)



(v)



b)

i) If  $y^2 = x^3(x-2)$

$$y^2 = x^4 - 2x^3$$

Diff w.r.t.  $x$

$$2y \frac{dy}{dx} = 4x^3 - 6x^2$$

$$y \frac{dy}{dx} = 2x^3 - 3x^2$$

$$\frac{dy}{dx} = \frac{x^2(2x-3)}{y}$$

$$\therefore \frac{dy}{dx} = \frac{x^2(2x-3)}{\sqrt{x^3(x-2)}}$$

$$= \frac{x^2(2x-3)}{x\sqrt{x(x-2)}}$$

$$= \frac{x(2x-3)}{\sqrt{x(x-2)}}$$

$$= \frac{\sqrt{x}(2x-3)}{\sqrt{x-2}}$$

$$= (2x-3)\sqrt{\frac{x}{x-2}}$$

(2)

ii) If  $y = x^3(x-2)$

$$= x^4 - 2x^3$$

$$y' = 4x^3 - 6x^2$$

$$\text{If } y' = 0, \quad 4x^3 - 6x^2 = 0$$

$$2x^2 - 3x = 0$$

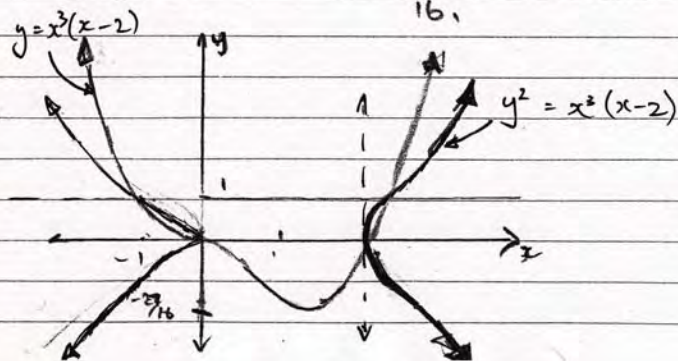
$$x^2(2x-3) = 0$$

$$x = 0 \text{ or } \frac{3}{2}$$

(3)

When  $y = \frac{3}{2}$ ,  $x = \frac{27}{8} (-\frac{1}{2})$

$$= -\frac{27}{16}$$



(iii) See graph.

(2)

QUESTION 3.

a) (i) Let  $P(x) = (x-\alpha)^r \cdot Q(x)$

$$\begin{aligned} P'(x) &= Q(x) \cdot r(x-\alpha)^{r-1} + (x-\alpha)^r \cdot Q'(x) \\ &= (x-\alpha)^{r-1} [r \cdot Q(x) + (x-\alpha) \cdot Q'(x)] \\ &= (x-\alpha)^{r-1} \cdot A(x) \text{ for some polynomial } A(x) \end{aligned}$$

$\therefore P'(x)$  has a root of multiplicity  $r-1$ . (2)

(ii)  $P(x) = x^4 - 5x^3 + 6x^2 + 4x - 8$

$$P'(x) = 4x^3 - 15x^2 + 12x + 4$$

$$P''(x) = 12x^2 - 30x + 12$$

$$= 6(2x^2 - 5x + 2)$$

$$= 6(2x-1)(x-2)$$

$P''(x)$  has roots  $\frac{1}{2}$  or 2

One of these is also a root of  $P'(x)$  &  $P(x)$

$$\text{Now } P'(2) = 4(2)^3 - 15(2)^2 + 12(2) + 4 = 0$$

$$= P(2) = (2)^4 - 5(2)^3 + 6(2)^2 + 4(2) - 8 = 0$$

$\therefore 2$  is the root of multiplicity 3.

$\therefore P(x) = (x-2)^3(x-b)$  for some value of  $b$

The product of the roots is  $(-2)^3 \cdot x - b = 8b$

$$\therefore 8b = -8$$

$$b = -1$$

$$\therefore P(x) = (x-2)^3(x+1)$$

(3)

b) (i) Since the coefficients of  $P(x)$  are real, if  $1+i$  is a root  $1-i$  is also a root i.e.  $1-i$  is a root of  $P(x)$

(1)

$$g) P(z) = z^3 - 3z^2 + 5z + 5$$

$$\text{Now } [z - (1+i)][z - (1-i)]$$

$$\alpha + \beta = 1+i + 1-i$$

$$= 2$$

$$\alpha\beta = (1+i)(1-i)$$

$$= 2$$

$$\therefore P(z) = (z^2 - 2z + 2)(az + b) \text{ for some values } a, b$$

$$= az^3 + bz^2 - 2az^2 - 2bz + 2az + 2b$$

$$= az^3 - (2a-b)z^2 + (2a-2b)z + 2b$$

$\therefore$  Equating coefficients

$$a = 2$$

$$\text{or } 2a - b = 3$$

$$4 - b = 3$$

$$\therefore b = 1$$

$$\therefore P(z) = (z^2 - 2z + 2)(2z + 1)$$

(3)

$$c) \text{ Let } P(x) = x^3 - 3x^2 + 4x - 2$$

$$P(1) = (1)^3 - 3(1)^2 + 4(1) - 2$$

$$= 0$$

$$\therefore x-1 \text{ is a factor}$$

$$x-1 \mid x^3 - 3x^2 + 4x - 2$$

$$x^3 - x^2$$

$$-2x^2 + 4x$$

$$-2x^2 + 2x$$

$$2x - 2$$

$$2x - 2$$

$$0$$

$$P(x) = (x-1)(x^2 - 2x + 2)$$

$$= (x-1)[(x^2 - 2x + 1) + 1]$$

$$= (x-1)[(x-1)^2 - i^2]$$

$$= (x-1)(x-1+i)(x-1-i)$$

(4)

$$d) \text{ Let the roots be } \alpha, \frac{1}{\alpha}, \beta, -\beta$$

$$\text{Now } \Sigma \alpha = \alpha + \frac{1}{\alpha} + \beta - \beta$$

$$\therefore \alpha + \frac{1}{\alpha} = \frac{10}{3}$$

$$3\alpha^2 + 3 = 10\alpha$$

$$3\alpha^2 - 10\alpha + 3 = 0$$

$$(3\alpha - 1)(\alpha - 3) = 0$$

$$\alpha = \frac{1}{3} \text{ or } 3$$

$$\text{Also } \Sigma \alpha\beta = \alpha \cdot \frac{1}{\alpha} \cdot \beta \cdot -\beta$$

$$\therefore -\beta^2 = -\frac{12}{3}$$

$$\beta^2 = 4$$

$$\beta = \pm 2$$

$$\therefore \text{Roots are } \frac{1}{3}, 3, 2, -2$$

$$\text{Also } \Sigma \alpha\beta = \frac{1}{3} \cdot 3 + \frac{1}{3} \cdot 2 + \frac{1}{3} \cdot -2 + 3 \cdot 2 + 3 \cdot -2 + 2 \cdot -2$$

$$= 1 + \frac{2}{3} - \frac{2}{3} + 6 - 6 - 4$$

$$= -3$$

$$\therefore -3 = p$$

$$p = -9^3$$

$$\therefore \Sigma \alpha p = \frac{1}{3} \cdot 3 \cdot 2 + \frac{1}{3} \cdot 3 \cdot -2 + \frac{1}{3} \cdot 2 \cdot 2 + 3 \cdot 2 \cdot -2$$

$$= 2 - 2 - \frac{4}{3} - 12$$

$$= -\frac{40}{3}$$

$$\therefore -\frac{40}{3} = -\frac{q}{3}$$

$$q = 40$$

(4)

$$\begin{aligned} \text{e) If } z &= \cos \theta + i \sin \theta \\ z^n &= (\cos \theta + i \sin \theta)^n \\ &= \cos n\theta + i \sin n\theta \end{aligned}$$

$$\begin{aligned} z^{-n} &= (\cos \theta + i \sin \theta)^{-n} \\ &= \cos(-n\theta) + i \sin(-n\theta) \\ &= \cos n\theta - i \sin n\theta \end{aligned}$$

$$\therefore z^n + z^{-n} = 2 \cos n\theta$$

$$\text{Now } z + \frac{1}{z} = 2 \cos \theta$$

$$\begin{aligned} \left(z + \frac{1}{z}\right)^5 &= (2 \cos \theta)^5 \\ &= 2^5 \cos^5 \theta \\ &= 32 \cos^5 \theta \end{aligned}$$

$$\begin{aligned} \text{But } \left(z + \frac{1}{z}\right)^5 &= z^5 + \binom{5}{1} z^4 \cdot \frac{1}{z} + \binom{5}{2} z^3 \cdot \frac{1}{z^2} + \binom{5}{3} z^2 \cdot \frac{1}{z^3} \\ &\quad + \binom{5}{4} z \cdot \frac{1}{z^4} + \frac{1}{z^5} \end{aligned}$$

$$= z^5 + 5z^3 + 10z + 10 \cdot \frac{1}{z} + 5 \cdot \frac{1}{z^3} + \frac{1}{z^5}$$

$$= \left(z^5 + \frac{1}{z^5}\right) + 5 \left(z^3 + \frac{1}{z^3}\right) + 10 \left(z + \frac{1}{z}\right)$$

$$\begin{aligned} &= 2 \cos 5\theta + 5 \cdot 2 \cos 3\theta + 10 \cdot 2 \cos \theta \\ &= 2 \cos 5\theta + 10 \cos 3\theta + 20 \cos \theta. \end{aligned}$$

(4)

Question 4.

$$\text{a) } \int \frac{e^x}{\sqrt{1-e^{2x}}} dx = \int \frac{f'(x)}{\sqrt{1-[f(x)]^2}} dx$$

$$\text{where } f(x) = e^x$$

$$= \sin^{-1} f(x) + C$$

$$= \sin^{-1} e^x + C.$$

(2)

$$\text{b) } \int_1^2 \frac{dx}{x\sqrt{2x-1}}$$

$$\text{If } u = (2x-1)^{1/2}$$

$$du = \frac{1}{2} (2x-1)^{-1/2} \cdot 2 dx$$

$$= \frac{dx}{\sqrt{2x-1}}$$

$$\text{If } u^2 = 2x-1$$

$$u^2 + 1 = 2x$$

$$x = \frac{1}{2} (1+u^2)$$

(4)

$$\text{When } x=1, u=1$$

$$x=2, u=\sqrt{3}$$

$$\begin{aligned} \therefore I &= \int_1^{\sqrt{3}} \frac{du}{\frac{1}{2}(1+u^2)} \\ &= 2 \left[ \tan^{-1} u \right]_1^{\sqrt{3}} \end{aligned}$$

$$= 2 \left( \tan^{-1} \sqrt{3} - \tan^{-1} 1 \right)$$

$$= 2 \left( \frac{\pi}{3} - \frac{\pi}{4} \right)$$

$$= \frac{\pi}{6}$$

$$c) (i) \frac{7}{2x^2+5x-3} = \frac{7}{(2x-1)(x+3)}$$

$$\text{If } \frac{7}{(x+3)(2x-1)} = \frac{a}{x+3} + \frac{b}{2x-1}$$

$$7 = a(2x-1) + b(x+3)$$

$$\text{Let } x = -\frac{1}{2}, \quad 7 = b \cdot 3\frac{1}{2}$$

$$\therefore b = 2$$

$$\text{Let } x = -3, \quad 7 = a \cdot -7$$

$$\therefore a = -1$$

$$\therefore \frac{7}{(x+3)(2x-1)} = \frac{-1}{x+3} + \frac{2}{2x-1}$$

(3)

$$(ii) \int \frac{7}{2x^2+5x-3} dx = \int \frac{-1}{x+3} + \frac{2}{2x-1} dx$$

$$= -\ln(x+3) + \ln(2x-1) + c$$

$$= \ln \left| \frac{2x-1}{x+3} \right| + c$$

(4)